THE KEY FORMULA IDENTIFYING A DIFFERENCE OF DIVISORS WEIGHTED BY ORDERS WITH THE DIVISOR OF THE TAME SYMBOL

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as two spans, lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Editorial note. The author's closing sentence at source line 4815, 'which is zero by the key Lemma \ref{lemma-milnor-gersten-low-degree}.', is retained verbatim at the end of the counted proof; the reference is kept as a tagged navigation pointer. The coefficient comparison that produces the displayed sum is carried out inside the span, so only the final citation is left to the reader.

Editorial note. Span note: the pool card recorded proof_lines 4760-4815, one line short of the closing \end{proof} at line 4816, which made the first assembly attempt fail with a runaway proof environment; the counted proof span is the real environment 4760-4816, taken from the pinned chapter file.

Allowed standard prerequisites (editorial).

- Well-definedness of the divisor of a meromorphic function. Divisors, label divisors-lemma-divisor-meromorphic-well-defined, invoked at proof line 4769. Justifies that replacing a trivialising section by a unit multiple changes the divisor by div(u|_{Z_i}), which is the computation in the first paragraph.
- Normalisation of the tame symbol and its bilinearity. Same chapter chow.tex, property item-normalization of the tame symbol, invoked at proof line 4775, together with bilinearity used at proof lines 4772-4774. Shows the two extra terms produced by the replacement s_i -> u s_i are equal, which is what allows the reduction a_i = b_i = 1.
- Vanishing of the sum of the orders of the tame symbol over the height one primes. Same chapter chow.tex, label lemma-milnor-gersten-low-degree, invoked at proof line 4815. Supplies the vanishing of the final sum of orders;

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the reduction of the coefficient comparison to that sum is carried out in the counted span.

Source reference. Original lemma 02SH: sources/stacks-chow/tags.txt, source lines 3660–3660. Navigation pointer for \ref{divisors-lemma-divisor-meromorphic-well-defined} used inside the counted span (line 4769); the argument that consumes it is retained in the span.

Source reference. Original lemma 0EAN: sources/stacks-chow/pinned-original-source.tex, source lines 790–790. Navigation pointer for \ref{item-normalization} used inside the counted span (line 4775); the argument that consumes it is retained in the span.

Source reference. Original lemma 0EAX: sources/stacks-chow/pinned-original-source.tex, source lines 1263–1263. Navigation pointer for \ref{lemma-milnor-gersten-low-degree} used inside the counted span (line 4815); the argument that consumes it is retained in the span.

Lemma 0.1 (Key formula)**.** *In the situation above the cycle*

$$\sum (Z_i \to X)_* \left(ord_{B_i}(f_i) div_{\mathcal{N}|_{Z_i}}(t_i|_{Z_i}) - ord_{B_i}(g_i) div_{\mathcal{L}|_{Z_i}}(s_i|_{Z_i}) \right)$$

is equal to the cycle

$$\sum (Z_i \to X)_* div(\partial_{B_i}(f_i, g_i))$$

Proof. First, let us examine what happens if we replace s_i by us_i for some unit u in B_i. Then f_i gets replaced by $u^{-1}f_i$. Thus the first part of the first expression of the lemma is unchanged and in the second part we add

$$-\mathrm{ord}_{B_i}(g_i)\mathrm{div}(u|_{Z_i})$$

(where $u|_{Z_i}$ is the image of u in the residue field) by Divisors, Lemma 02SH and in the second expression we add

$$\mathrm{div}(\partial_{B_i}(u^{-1}, g_i))$$

by bi-linearity of the tame symbol. These terms agree by property (0EAN) of the tame symbol.

Let $Z \subset X$ be an irreducible closed with $\dim_\delta(Z) = n - 2$. To show that the coefficients of Z of the two cycles of the lemma is the same, we may do a replacement $s_i \mapsto us_i$ as in the previous paragraph. In exactly the same way one shows that we may do a replacement $t_i \mapsto vt_i$ for some unit v of B_i.

Since we are proving the equality of cycles we may argue one coefficient at a time. Thus we choose an irreducible closed $Z \subset X$ with $\dim_\delta(Z) = n - 2$ and compare coefficients. Let $\xi \in Z$ be the generic point and set $A = \mathcal{O}_{X,\xi}$. This is a Noetherian local domain of dimension 2. Choose generators σ and τ for $\mathcal{L}_\xi$ and $\mathcal{N}_\xi$. After shrinking X, we may and do assume σ and τ define trivializations of the invertible sheaves $\mathcal{L}$ and $\mathcal{N}$ over all of X. Because Z_i is locally finite after shrinking X we may assume $Z \subset Z_i$ for all $i \in I$ and that I is finite. Then ξ_i corresponds to a prime $\mathfrak{q}_i \subset A$ of height 1. We may write $s_i = a_i\sigma$ and $t_i = b_i\tau$ for some a_i and b_i units in $A_{\mathfrak{q}_i}$. By the remarks above, it suffices to prove the lemma when $a_i = b_i = 1$ for all i.

Assume $a_i = b_i = 1$ for all i. Then the first expression of the lemma is zero, because we choose σ and τ to be trivializing sections. Write $s = f\sigma$ and $t = g\tau$ with f and g in the fraction field of A. By the previous paragraph we have reduced to the case $f_i = f$ and $g_i = g$ for all i. Moreover, for a height 1 prime $\mathfrak{q}$ of A which is not in $\{\mathfrak{q}_i\}$ we have that both f and g are units in $A_\mathfrak{q}$ (by our choice of the family $\{Z_i\}$

in the discussion preceding the lemma). Thus the coefficient of Z in the second expression of the lemma is

$$\sum\nolimits_i \mathrm{ord}_{A/\mathfrak{q}_i}(\partial_{B_i}(f,g))$$

which is zero by the key Lemma 0EAX. □